Lodestar

Bankable demand forecasts that tell mini-grid developers and their lenders how much power a village will use, before anyone builds.

BRIEF
03 · Predict mini-grid demand before building

TRACK
2 · Digital Solutions & AI

TEAM
Tushar Agarwal · product, data and engineering

HOW THIS PITCH IS SCORED

Impact	25%
Innovation	25%
Sustainability	25%
Feasibility	15%
Scalability	10%

IMPACT · 25% OF SCORE

Every mini-grid is sized on a guess about demand

THE PROBLEM

490 million people need the 217,000 mini-grids still to be built (ESMAP). Each is sized before anyone connects. Too big, and tariffs price villagers out. Too small, and evenings go dark.

"Survey-based estimates miss individual customers' demand by 65%." (CrossBoundary Innovation Lab)

5 of 6

years where real metered demand landed inside our forecast range (DR Congo, forecast without the site's meters)

+37% → +6%

oversizing at a real site: today's desk estimate vs Lodestar's plan

−48%

cost of power when built for likely demand, not the optimistic case (20 villages)

Bottom line: Tested on real meters: demand landed in our range 5 of 6 times, and right-sizing roughly halves the cost of power.

INNOVATION · 25% OF SCORE

We forecast demand as a range lenders can bank on

TODAY

One day-one number from a survey or benchmark: no range, no growth, never checked against meters.

OUR APPROACH

A probabilistic range, month by month: a safe level demand beats 9 times in 10, the likely level, and the upside.

ROOT CAUSE WE ADDRESS

No trusted demand number before building, so sites are mis-sized or never financed.

HOW IT WORKS

1. Click a village: satellite buildings and OpenStreetMap count customers
2. 20,000 growth scenarios give a five-year range, in plain words
3. Size the system, stress-test the loan; operators' meters sharpen it

Bottom line: A range instead of a guess, with growth over time and a lender view, tested on real meters.

SUSTAINABILITY & AFFORDABILITY · 25% OF SCORE

Villagers pay nothing; developers and funders do

COST TO THE USER

$0

for households. A developer pays $300 per site report: 0.05% of a ~$585k mini-grid.

BREAK-EVEN

6–8 customers

WHO PAYS

Developers: $300 per site report. Funders (RBF programmes, DFIs): $12k/yr to screen portfolios. Lenders: later, per loan file.

HOW IT PAYS FOR ITSELF

Costs ~$60k/yr (two analysts, one server). Covered by 3 funder subscriptions + ~80 reports, e.g. 3–5 developers.

MARKET

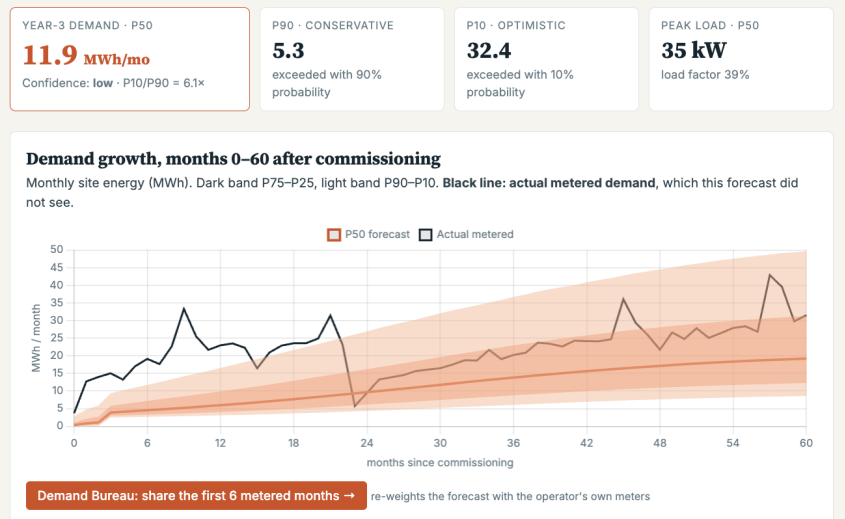
AMDA members plan 6,116 sites (≈$1.8M in reports). Nigeria's DARES alone: 1,000+. Needed by 2030: 217,000.

ENVIRONMENTAL BENEFIT

No stranded panels and batteries (desk plan oversized a real site by 61 kWp, ours by 9) and less backup diesel.

Bottom line: Villagers pay nothing. 6–8 paying customers cover costs, in a pipeline of thousands of planned sites.

Live and tested; next, a 10-site pilot in Nigeria



EVIDENCE IT WORKS

- Real meters: demand in range 5 of 6 years
- Exports to HOMER; open data and licences
- Risk: few public meters → backtest operators first

PATH TO SCALE

PILOT

Nigeria: 10 sites with DARES developers

EXPAND

Score on meters; one funder; Kenya next

SCALE

5 countries: ~$30k/yr per country (est.), compute ~free

OUR ASK

2–3 operators' past meter data, and one RBF programme as pilot partner. They get back: tighter forecasts (cheaper debt), peer benchmarks, free reports.

Bottom line: Live at lodestar-utfz.onrender.com. Partners who share meter data get tighter forecasts and cheaper debt in return.